HONORS ALGEBRA MIDTERM

. . . when you apply commutative algebra, you can come to conclusions even if you don't know what you are doing at each step along the way. — Heisuke Hironaka

All[1] rings in this exam are **commutative with identity**. You are required to show **detailed** work for all problems. You may assume, if necessary, any results stated in lectures or assertions from homework problems. The exam is worth a total of **100 points**.

Problem 1 [15pt]. — For each part, find an example of a module satisfying the stated property (also state explicitly the base ring for this module). You do **not** need to justify your answers.

(1) A projective module which is not free.
(2) A divisible module which is not injective.
(3) A torsion-free module which is not flat.
(4) A flat module which is not projective.
(5) A direct product of projective modules which is not projective.

Problem 2 [10pt]. — Consider the prime ideal $\mathfrak{p} = \left(2, 1+\sqrt{5}\right)$ of $A = \mathbb{Z}\left[\sqrt{5}\right]$. Is $\mathfrak{p}$ a projective A-module? Provide proof.

Problem 3 [10pt]. — Let $T \in \mathrm{M}_{m\times n}(\mathbb{Z})$ be an integral matrix of rank m.
Prove that the following statements are equivalent:

(1) The rows of T can be extended to a $\mathbb{Z}$-basis of $\mathbb{Z}^n$.
(2) The greatest common divisor of the $m \times m$ minors of T is 1.
(3) Any $\mathbb{Z}$-linear functional on R can be extended to a $\mathbb{Z}$-linear functional on $\mathbb{Z}^n$ where R is the $\mathbb{Z}$-submodule of $\mathbb{Z}^n$ generated by the rows of T.

Problem 4 [10pt]. — Let k be a field and let x, y, t be indeterminates. Show that the k-algebra homomorphism

$$f : k[x, y] \to k[x, t]$$

mapping $x \mapsto x$ and $y \mapsto xt$ is **not** flat.

Problem 5 [10pt]. — Let A be a ring and let A-Mod denote the category of A-modules.

(1) Show that a functor $F : A\text{-Mod} \to \mathrm{Set}$ is representable if and only if F admits a left adjoint.
(2) Let $F : A\text{-Mod} \to \mathrm{Set}$ be the forgetful functor. Classify all natural transformations from F to F. Show your work.

Date: April 27th, 2025.

[1]Except for the matrix ring $\mathrm{M}_{m\times n}(\mathbb{Z})$ appearing in Problem 3, when $m = n > 1$.

Problem 6 [10pt]. — Let A be an integral domain such that every prime ideal of A is principal. Show that A is a principal ideal domain (PID).

Hint: If $\mathfrak{a} \subseteq \mathfrak{b}$ with $\mathfrak{b}$ principal, then $\mathfrak{a} = (\mathfrak{a} : \mathfrak{b})\mathfrak{b}$.

Problem 7 [10pt]. — Let A be a ring and let $\mathfrak{a} \subseteq A$ be an ideal.

(1) Show that $A/\mathfrak{a}$ is a flat A-module if and only if for every ideal $\mathfrak{b}$ of A, we have $\mathfrak{a} \cap \mathfrak{b} = \mathfrak{ab}$.
(2) Let $(A, \mathfrak{m})$ be a local ring with residue field k. Use (1) to show that if k is a flat A-module, then $\mathfrak{m} = 0$.

Problem 8 [10pt]. — Let A be a ring, M an A-module. Prove the following statements:

(1) If $A \oplus M \cong A$, then $M = 0$.
(2) If $A \oplus M \cong A^2$, then $M \cong A$.

Problem 9 [15pt]. — Let A be a ring and let $M \subseteq E$ be A-modules.

We say that M is an **essential submodule** of E (or that E is an **essential extension** of M) if any nonzero submodule of E intersects M nontrivially. That is, $N \neq 0 \Rightarrow N \cap M \neq 0$ for all $N \subseteq E$.

Prove the following statements:

(1) An A-module M is injective if and only if it has no proper essential extensions (i.e. if $M \subseteq E$ is essential, then $M = E$).
(2) Given a pair of A-modules $M \subseteq Q$, there exists a maximal submodule $E \subseteq Q$ containing M such that $M \subseteq E$ is essential.
(3) If the module Q in (2) is injective, then so is E.
(4) There is, up to isomorphism, a unique essential extension E of M that is an injective A-module. This E is called the **injective envelope** of M.
(5) Compute the injective envelopes of the $\mathbb{Z}$-modules $\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}$ ($n \geqslant 2$). Provide proof.

For each step, you may assume the results of the previous steps without proof.